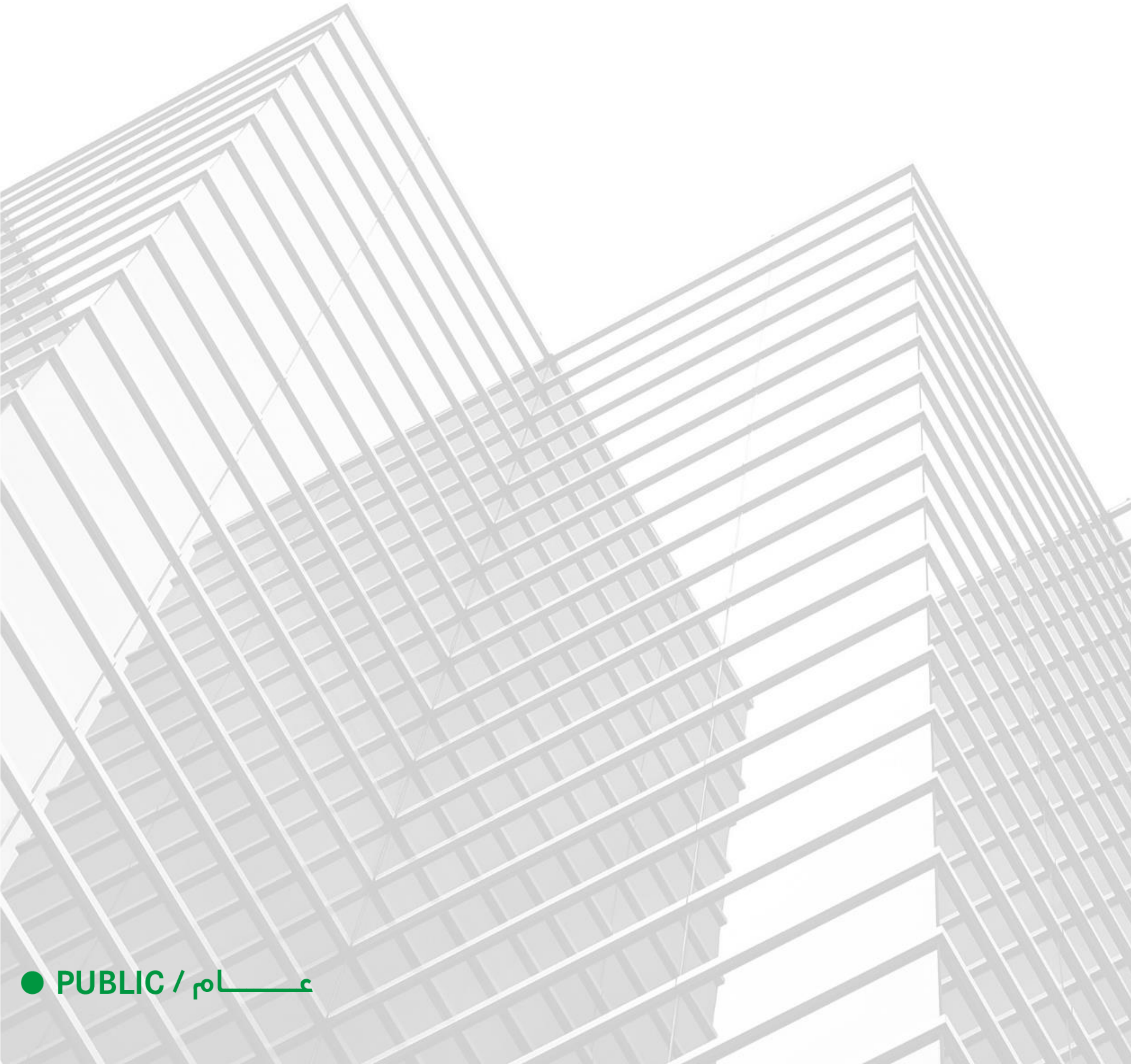
دائرة الصحة
DEPARTMENT OF HEALTH

GUIDELINES ON REAL-WORLD DATA/REAL-WORLD EVIDENCE-BASED CLINICAL RESEARCH



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Contact	Dmp@doh.gov.ae, RIC@doh.gov.ae

1. Guideline Purpose and Brief

The purpose of this guideline is to provide guidance on the Department of Health (DOH), Abu Dhabi requirements that apply when assessing clinical research using real-world data (RWD) or generating real-world evidence (RWE). Clinical Research conducted using RWD/RWE are subject to requirements outlined in relevant DOH standards and guidelines for clinical research, such as “DOH Standard on Human Subject Research” and “DOH guidelines for conducting clinical trials with investigational products and medical devices.” This document outlines specific additional requirements for RWD studies that aim to generate RWE to support regulatory, clinical or and reimbursement decisions. It also provides an overview on common types of RWD studies and data requirements.

The guideline applies to all sponsor/clinical research organizations, DOH-licensed healthcare entities, public and private institutions and researchers, seeking to conduct clinical research using RWD and RWE.

2. Definitions and Abbreviations

No.	Term / Abbreviation	Definition
2.1	Real-world data / RWD	The data related to patient health status and/or the delivery of routine health care which is collected outside of a clinical trial, for example data from electronic health records, patient-generated health data (including patient reported outcomes), administrative claims and billing activities, medical product/device registries, and data generated from other sources that could inform on health status, such as mobile devices.
2.2	Real-world evidence / RWE	The clinical evidence regarding the usage and potential benefits or risks of medical product derived from analysis of RWD (for example from randomized trials, pragmatic trials, and observational studies).
2.3	Electronic health records/ EHR	An individual patient record contained within an Electronic Health Record (EHR) system. A typical individual EHR may include a patient’s medical history, diagnoses, treatment plans, immunization dates, allergies, radiology images, pharmacy records, and laboratory and test results.
2.4	Registries	An organized system that collects clinical and other data in a standardized format for a population defined by a particular disease, condition, or exposure (diseases registries, health services registries, product registries)
2.5	Common Data Model / CDM	Standardizes a variety of electronic health care data sources into a common format to ensure interoperability across all sites providing data
2.6	Administrative claims data	Claims data arising from a person's use of the healthcare system (and reimbursement of healthcare providers for that care)
2.7	Digital Health Technology / DHT	A system that uses computing platforms, connectivity, software, and sensors for health care and its related uses. These technologies span a wide range of uses, from general wellness applications to medical devices. They include technologies intended for use as a medical product, in a medical product, or as an adjunct to other medical products (devices, drugs, and biologics). They may also be used to develop or study medical products.

2.8	Reliability of data	Includes data accuracy, completeness, provenance, and traceability
2.9	Relevance of data	Relevance includes the availability, consistency of key data elements (exposure, outcomes, and covariates) and sufficient numbers of representative patients for the study.
2.10	Provenance	An audit trail that “accounts for the origin of a piece of data (in a database, document or repository) together with an explanation of how and why it got to the present place”.
2.11	“Fit for purpose” data	Data which are considered to be used as RWD based on its relevance (availability, consistency of key data elements and a sufficient number of representative patients for the study), and reliability (data accuracy, completeness, provenance, and traceability)
2.12	Pharmacoepidemiology	The study of interactions between drugs and human populations, investigating, in real conditions of life, benefits, risks and use of drugs through application of epidemiologic methods. [17]
2.13	Pharmacoepidemiologic studies	Studies with various designs, including cohort (prospective or retrospective), case-control, nested case-control, cross-sectional, or other models, and the results of such studies may be used to characterize one or more safety signals associated with a product, or may examine the natural history of a disease or drug utilization patterns. These studies can allow for the estimation of the relative risk of an outcome associated with a product, and some (e.g., cohort studies) can also provide estimates of risk (incidence rate) for an adverse event.
2.14	Good Pharmacoepidemiology practice / GPP	Essential practices and procedures that should be considered to help ensure the quality and integrity of pharmacoepidemiologic research, and to provide adequate documentation of research methods and results. The GPP do not prescribe specific research methods, nor will adherence to guidelines to guarantee valid research. The GPP are intended to apply broadly to all types of pharmacoepidemiologic research, including feasibility assessments, validation studies, descriptive studies, as well as etiologic investigations, and all of their related activities from design through publication.
2.15	Metadata	A set of data that describes and gives information about a dataset, including: i. the generation, location, and ownership of the data set; key variables; and the format (coding, structured versus not) in which the data are collected. ii. the provenance and time span of the data, clearly documenting the input, systems, and processes that define data of interest. iii. the storage, handling processes, access, and governance of data.

3 Guideline Content

3.1 Introduction to RWD/RWE

Real-world data is continuously generated from many sources. Diagram 1 below presents the provenance and flow of data in real-world applications. RWD is processed using appropriate collection, standardization, and anonymization methods through data-sharing platforms. It forms a solid trunk from which analysis algorithms and visualization tools are applied to create RWE that branches out across various research and applications.

Each data source should be evaluated to determine whether the available information is appropriate for addressing a specific study hypothesis. Given that existing real-world data is not collected primarily for research, it is important to understand its potential limitations when used for that purpose.

The life cycle of data is an integral part of managing RWD and transforming it into RWE. The life cycle involves multiple phases: data accrual from the original source; curation of data to the clinical data repository; transformation and de-identification of data where necessary, creation of a data warehouse; and production of a study-specific dataset for analysis.

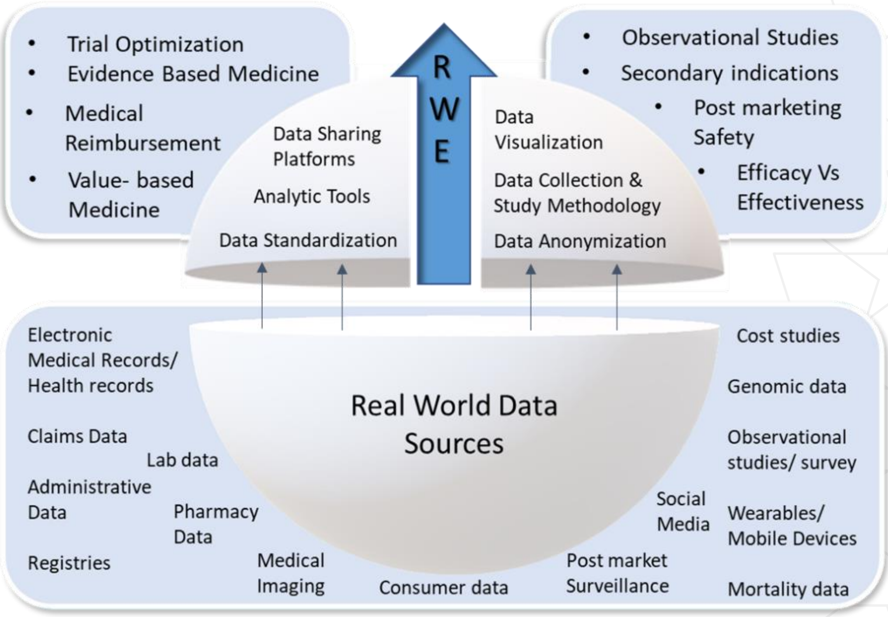

Diagram 1: Flow of Real-World Data

Real World Data/Real World Evidence has several applications in clinical research. RWD may be used in planning clinical trials for example to assess trial feasibility, and support site selection.

RWD may also be used in randomized trials with pragmatic, non- randomized, single group interventional studies and non-interventional, observational studies. RWD studies may be retrospective using secondary data sources or prospective using primary data collection methods.

Use of RWD sources in clinical studies has the potential to increase the speed and reduce the cost of the process of regulatory approval and monitoring of medical products, consequently impacting the public health. RWE may be more representative of the true effects of a particular treatment and more generalizable than data from controlled clinical trials. In addition to medicines evaluation, RWE serves public health needs in other domains such as health technology assessment, disease epidemiology and health economics.

RWE has been widely used in the context of post marketing surveillance to monitor drug safety and detect adverse events. RWE is particularly effective in situations where the outcome of interest is rare, in cases where a very long follow-up period is required to assess the health outcomes, or when it is difficult to perform randomized controlled trials (RCTs), such as in pediatric or pregnant populations.

Pharmacoepidemiology, the study of uses, risks and therapeutic effects of health products in large populations using epidemiological methods,[17] is the core discipline that supports the generation and assessment of valid and reliable RWE. Adherence to internationally recognized Good Pharmacoepidemiology Practices (GPP) guidelines such as "ENCePP Guide on Methodological Standards in Pharmacoepidemiology"[5] is recommended when conducting RWE studies.

The “STaRT- RWE: structured template for planning and reporting on the implementation of real-world evidence studies” [12] also provides designing and reporting guidance to conduct reproducible studies of safety and effectiveness of medical products. This template facilitates transparent communication of RWE study methodology allowing validity assessment and evidence synthesis.

In addition, the ENCePP checklist for Study Protocols available at appendix 1 provides a list of important elements to be considered when designing pharmacoepidemiology study protocols using RWD. High quality RWD studies using “fit for purpose” data and appropriate methodologies result in reliable and reproducible RWE which can be used to support the development, access and use of safe and effective medical products.

This guideline presents general points to be considered when planning to conduct clinical research using RWD to generate RWE which can be used in regulatory and clinical decision-making. RWD and RWE research is an evolving field with new data sources and methodologies being explored and recognized. The rapid advancements in healthcare digitalization have a strong influence on the implementation and advancement of real-world data research.

It is important to note that compliance to this guideline or any international best practice guideline in the conduct of RWE studies does not guarantee a favorable decision towards a medical product, as decision-taking depends on overall assessment of study results, data utilized, methodologies employed and other relevant factors.

3.2 Introduction to Data Requirements

3.2.1 Data Sources

i. Data Variation across different healthcare systems

When information on data sources is reviewed, the differences in the practice of medicine around the world, and between health care systems should be considered, as they will differ in a range of characteristics, such as age, socioeconomic status, health conditions, risk factors, and other potential confounders. There are various factors in health care systems and insurance programs, such as medication tiering, formulary decisions, and patient coverage which will influence the degree to which patients on a given therapy in one health care system might differ in disease severity, or other disease characteristics, from patients on the same therapy in another health care system.

ii. Identification, validation, and relevance of data

When assessing whether the data sources cover all populations relevant to the study and if those sources are to be used to examine the study hypothesis, the following items should be considered:

i. The reason for selecting the particular data sources to address the specific hypotheses.

ii. Background information about the health care system, including (if available) any specified method of diagnosis and preferred treatments for the disease of interest, and the degree to which such information is collected and validated in the proposed data sources.

iii. A description of prescribing and use practices in the health care system (if available), including for approved indications, formulations, and doses.

iii. Study Population Selection

In addition to the minimal data requirement to ensure ‘need-to-know’, the protocol should include a detailed description of methods for determining how inclusion and exclusion criteria (e.g., demographic factors, medical condition, disease status, severity, biomarkers) will be implemented to identify appropriate patients meeting these criteria from the data source. The protocol should address the completeness and accuracy of the information collected in the proposed data source to fulfill the inclusion and exclusion criteria. Key variables used to select the study population should be validated. In certain circumstances, key variables (e.g., gestational age for pregnancy studies) required to fulfill the inclusion and exclusion criteria may be generated by the health care provider using information available at the point of care. If such data are used, the protocol should describe the source of information and the methods health care providers use to generate the data (if known).

Selection of study population should be appropriate in terms of size, coverage and representativeness.

3.2.2 Approaches of data collection

There are two main approaches for data collection: collection of data specifically for a particular study ('primary data collection') or use of data already collected for purposes other than research, e.g. as part of administrative records of patient health care ('secondary use of data').

Secondary use of data has become a common approach in pharmacoepidemiology globally due to the increasing availability of electronic healthcare records, administrative claims data and other already existing data sources, and due to its increased efficiency and lower cost in terms of timely and rapid data generation.

In addition, networking between international centers active in the pharmacoepidemiology and pharmacovigilance fields is rapidly changing the landscape of drug safety and effectiveness research, both in terms of availability of networks of data sources and networks of researchers who can contribute to a particular study with a particular data source.

A. Primary data collection

The methodological aspects of primary data collection studies involve different study designs based on prospective primary data collection such as cross-sectional study, prospective cohort study, registries, active surveillance, and surveys.

Studies using hospital or community-based primary data collection have allowed the evaluation of drug-disease associations for rare complex conditions that require very large source populations and in-depth case assessment by clinical experts.

Data can be collected using paper, electronic case report forms or, increasingly, smartphone or web applications provided to patients.

i. Non-interventional prospective cohort study with primary data collection

The specifics of design of this type of study is presented under observational studies section 3.2.1.

ii. Cross-sectional studies – Surveys

A survey is the collection of data on knowledge, attitudes, behavior, practices, opinions, beliefs or feelings of selected groups of individuals from a specific sampling frame, by asking them questions in person or by post, phone or online. They generally have a cross-sectional design, but repeated measures over time may be performed for the assessment of trends. This type of research allows for a variety of methods to recruit participants, collect data and utilize various instruments. An important part of surveys is the selection of the sampling method, such as clustered random sample, where the target population has to be considered.

iii. Registries

Although existing patient registries may be considered as secondary data source for conducting studies, as explained under secondary use of data, registries can also collect data prospectively and be considered as a primary data collection method.

B. Secondary use of data

This is utilization of data already collected for another purpose (e.g. as part of administrative records of patient health care) and that can be linked to prospectively collected medical and non-medical data. Electronic healthcare database (e.g. administrative claims databases, electronic medical records / electronic health records) and patient registries are examples of sources of data that can be leveraged as secondary data for pharmacoepidemiology studies.

i. Patient registries

This is an organized system that collects data and information on a group of patients defined by a particular disease, condition or exposure, and that serves a pre-determined scientific, clinical and/or public health (policy) purpose. A registry-based study is the investigation of a research question using the data collection infrastructure or patient population of one or more existing or new patient registries. A registry-based study may be a non-interventional trial/study or a clinical trial/study. A patient registry should be considered as an infrastructure for the standardized recording of data from routine clinical practice on individual patients identified by a characteristic or an event – disease registry, pregnancy registry, birth defect registry, a molecular or a genomic feature or any other patient characteristics, or an encounter with a particular healthcare service.

To address a specific study objective, data from a patient registry may need to be enriched with additional information on outcomes, lifestyle data, immunization or mortality information which can be obtained through linkage with other existing databases.

ii. Spontaneous reports

Spontaneous reports of adverse drug effects remain a cornerstone of pharmacovigilance and are collected from a variety of sources, including healthcare providers, national authorities, pharmaceutical companies, medical literature and more recently directly from patients. The strength of spontaneous reporting systems is that they cover all types of authorized drugs used in any setting (primary, secondary or specialized healthcare). In addition to this, the reporting systems are built to obtain information specifically on potential adverse drug reactions. The information collected in spontaneous reports reflects a clinical event that has been attributed to the use of one or more suspected drugs.

iii. Other digital data sources

Technological advances have increased the range of data sources that can be used to complement traditional ones and may provide compelling insights into or relevant to effectiveness and safety of health interventions. Digital sources are increasingly used in medicines and their risk minimization measures, benefit-risk communications and related stakeholder engagement. Such data include those from digital media that exist in a computer-readable format and can be extracted from websites, web pages, blogs, vlogs, social networking sites, internet forums, chat rooms and health portals.

Digital data sources can also contain subjective patient-reported outcomes and feedback on quality of life (QoL) along with sentiment that can be analyzed using natural language processing or sentiment analysis to obtain insights that might be found in other data sources.

A recent addition to the digital media data is biomedical data collected through wearable technology (e.g., heart rate, physical activity and sleep pattern, dietary patterns). These data are unsolicited and generated in real time.

iv. Multi-database studies

Pooling data across different databases affords insight into the generalizability of the results and may improve precision. Globally, a growing number of studies use data from networks of databases, often from different countries. Some of these networks are based on long-term contracts with selected partners and are very well structured while others are looser collaborations based on an open community principle.

Advantages of these type of studies include increased study population size, reduced time to get desired sample size, obtaining knowledge on generalizability of results and consistency of information across countries and increased amount of information on a specific issue addressed in different databases.

These studies involve experts from various countries addressing case definitions, terminologies, coding in databases and research practices providing opportunities to increase consistency of results of observational studies.

3.1.2 Requirements to data capture:

i. Enrollment and Comprehensive Capture of Care

Continuity of coverage (enrollment and disenrollment) should be addressed when using EHR, and medical claims data sources, given that patients often enroll and disenroll in different health plans when they experience changes in employment or other life circumstances.

ii. Data Linkage and Synthesis

Data linkages can be used to increase the breadth and depth of data on individual patients over time and provide additional data for validation purposes. When the study involves establishing new data linkages between internal data sources or external data sources (e.g. vital records, disease and product registries, biobank data), the protocol has to include description of data source, the information that will be obtained, linkage methods, and the accuracy and completeness of data linkages over time. When the study involves generating additional data (e.g. interviews, mail surveys, computerized or mobile-application questionnaires, measurements through digital health technologies), the protocol should describe the methods of data collection and the methods of integrating the collected data with the electronic health care data.

iii. Distributed Data Networks

Distributed data networks (or systems) of EHRs and medical claims data systems, often combined with the use of Common Data Models (CDMs), have been increasingly used globally for medical product safety surveillance and research purposes. In these networks in which data from multiple sites are transformed into a single CDM, the primary benefit is the ability to execute an identical query (without any or substantial modifications) on multiple datasets. In some distributed data networks, queries can be run simultaneously at all network sites or at each site asynchronously, with results combined at a coordinating center for return to the end user. Distributed data networks are typically comprised of EHR, medical claim, or registry data.

iv. Computable Phenotypes

They facilitate the identification of similar patient populations and enable efficient selection of populations for large-scale clinical studies across multiple health care systems. A computable phenotype definition should include metadata and supporting information about the definition, its intended use, the clinical rationale or research justification for the definition, and data assessing validation in various health care settings.[19] The computable phenotype definition, composed of data elements and phenotype algorithm, should be described in the protocol and study report and should also be available in a computer-processable format. Clinical validation of the computable phenotype definition should be described in the protocol and study report.

v. Unstructured Data

Large amounts of clinical data are unstructured data within EHRs, either as free text data fields (such as physician notes) or as other non-standardized information in computer documents (such as PDF-based radiology reports).

Technology advances of artificial intelligence (AI) may permit processing of such data with advances of natural language processing, machine learning and deep learning to extract data elements from unstructured text in addition to structured fields in EHRs; developed computer algorithms that identify outcomes; or evaluate laboratory images results. These methods are computer-assisted and require human-aided curation and decision – making. In case AI is proposed to be used or other derivation methods – the study protocol should specify the assumptions and parameters of the computer algorithms used, data sources, supervised or unsupervised algorithm use and the metrics associated with the validation of the method. These specifications have to be documented in the study protocol.

3.1.3 Data quality

The quality of the source data includes its accuracy, validity, variability, reliability, and provenance. Quality assurance of data from the origin of the source database is also important due to possibility of transformation and manipulation during data processing. Managing of data quality refers to the user's capability to define the data's provenance, mechanisms of how the data links are made, how the discrepancies are ruled and how the descriptions of any limitations or considerations associated with the data are made. The presence of Metadata of the data sources used is also necessary to ensure quality and appropriateness of the data source.

RWD extracted from a database, or other data sources must demonstrate sufficient quality for the intended use. It is recommended to follow the "Guidelines for Good Database Selection and use in Pharmacoepidemiology Research"[11] prepared by members of the International Society of Pharmacoepidemiology (ISPE) to support obtaining a high confidence level in the data used. The guidelines include principles for the selection of a database, use of multiple resources, extraction and analysis of the study population, privacy and security, quality and validation procedures, and reliable documentation. [11]

The study protocol must include a description of the tools and methods for selection, extraction, transfer, and secure handling of the data and their validation. Acceptable processing is expected to be in place for any additional data collected outside of the primary source database.

3.1.4 Digital health technologies

Digital health technologies are used to gather health-related data. These include applications, sensors, wearables, and other technologies, such as ingestible devices and implants, which might be used to collect RWD as part of routine clinical care for patient-reported outcomes or home-based measurements. Other sources of RWD might utilize devices that capture indicators of function. These might be used as supportive evidence of the safety or efficacy of a treatment. Both the device and any tool (such as a questionnaire) used in data capturing have to be suitability validated for the measurements required. The relevance, objectivity, and practicality of measurements should be considered, considering the user's disease, age, and potential functional abilities. The devices may be regulated as medical devices, and applicable laws and regulations on the territory of Abu Dhabi are to be applied.

3.2 Study designs and characteristics of different types of RWD/RWE studies

3.2.1 Observational studies (Non-interventional studies – NIS)

Refer to relevant DOH Guidelines for conducting clinical trials for information on basic requirements for non-interventional observational research in Abu Dhabi. Below is a brief outline on study designs commonly used in RWD studies.

i. Cohort studies

The investigator identifies a population at risk or the outcome of interest, defines one or more groups of people (study cohort) who are free of disease and differ according to their extent of exposure, and follows them over time to observe the occurrence of the disease in the exposed and unexposed cohorts.

In a cohort study, a population-at-risk for an event of interest is followed over time for the occurrence of that event. Information on exposure status is known throughout the follow-up period for each study participant. A study participant might be exposed to a medicinal product at one time during follow-up, but unexposed at another time point. Since the population exposure during follow-up is known, incidence rates can be calculated.

In many cohort studies (prospective and retrospective) involving exposure to medicinal product(s), comparison cohorts of interest are selected on the basis of medication use and followed over time. Cohort studies are useful when there is a need to know the incidence rates of adverse events in addition to the relative risks of adverse events. They are also useful for the evaluation of multiple adverse events within the same study. However, it may be difficult to recruit sufficient numbers of patients who are exposed to a product of interest (such as an orphan medicinal product) or to study very rare outcomes. The identification of patients for cohort studies may come from large automated databases or from data collected specifically for the study at hand. In addition, cohort studies may be used to examine safety concerns in special populations (older persons, children, patients with comorbid conditions, pregnant women) through over-sampling of these patients or by stratifying the cohort if sufficient numbers of patients exist.

ii. Case-control studies

In these studies, cases of disease (or events) are identified and patients from the source population who do not have the disease or event of interest at the time of selection are selected as controls. Matching can be done here to control for possible confounders. The odds of exposure are then compared between the two groups. Patients may be identified from an existing database, using a field study approach, in which data are collected specifically for the purpose of such kind of study or if the focus is on special populations, the cases and controls may be stratified according to the population of interest.

These studies are particularly useful when the goal is to investigate whether there is an association between a medicinal product (or products) and one specific rare adverse event, as well as to identify multiple risk factors for adverse events. If all cases of interest (or a well-defined fraction of cases) are captured and the fraction of controls from the source population is known, a case-control study may provide the absolute incidence rate of the event.

Another subtype is “nested case-control study” which has been designated as the study in which the control sampling is density-based. Another variant is in which the control sampling is performed on those persons who make up the source population regardless of the duration of time they may have contributed to it. A case-control approach could also be set up as a permanent scheme to identify and quantify risks (case-control surveillance).

iii. Case-only designs (self-controlled)

These studies have been proposed to assess the association between intermittent exposures and short-term events, including the self-controlled case-series, the case-crossover and the case-time control studies. In these designs, only cases are used and the control information is obtained from person-time experience of the cases themselves.

One of the important strengths of these designs is that confounding variables that do not change over time within individuals are automatically matched. Case-only designs cannot be used under all circumstances, for instance when the exact date of disease onset is difficult to establish or when evaluating chronic exposures.

iv. Cross-sectional studies

Data collected on a population of patients at a single point in time (or interval of time) regardless of exposure or disease status constitute a cross-sectional study.

The main purpose of these studies is to gather data for surveys or for ecological analyses. A drawback of cross-sectional studies is that the temporal relationship between exposure and outcome cannot be directly addressed, which limits its use for etiologic research unless the exposure does change over time. Also, they are used to examine the prevalence of a disease at one point in time or to examine trends over time where data for serial time-points can be captured, and to examine the crude association between exposure and outcome in ecological analyses.

Surveys distributed at one point in time are also used to collect patient reported outcomes (PROs) or quality of life (QoL) questionnaires.

3.2.2 EHR/Active surveillance

i. Registry-based studies

These studies select participants, based on diagnosis of a disease, prescription of a medicinal product, or both. The choice of such kind of population and the design of the registry usually is driven by its objective(s) in terms of outcomes to be measured and analyses and comparisons to be performed. Registry-based studies are applicable to rare disease, rare exposure or special population.

In many cases, registries can be enriched with data on outcomes, confounding variables and effect modifiers obtained from a linkage to an existing database such as national cancer registries, prescription databases, mortality records or other relevant databases. Depending on their objective, registries may provide data on patient, disease and treatment outcomes, and of their determinants.

Data on outcomes may include data on patient-reported outcomes, clinical conditions, medicines utilization patterns and safety and effectiveness. In some occasions, registries may be the only opportunity to provide insight into efficacy aspects of a medicinal product, but they are not normally used to demonstrate efficacy.

After performance of clinical trials (RCTs), patient registries may be useful to study effectiveness in heterogeneous populations, effect modifiers, such as doses that have been prescribed by physicians and that may differ from those used in RCTs, patient sub-groups defined by variables such as age, co-morbidities, use of concomitant medication or genetic factors, or factors related to a defined country or healthcare system.

Where adequate data are already available or can be collected, patient registries may be used to compare risks of outcomes between different groups. Patient registries may address exposure to medicinal products in specific populations, such as pregnant women. Patients may be followed over time and included in a cohort study to collect data on adverse events using standardized questionnaires. Simple cohort studies may measure incidence, but, without a comparison group, cannot evaluate any association between exposures and outcomes.

Registries may have advantages over other RWD sources, given that registries collect structured and predetermined data elements and can offer longitudinal, curated data about a defined population of patients and their corresponding disease course, complications, and medical care. In addition, registries can systematically collect patient-reported data that medical claims datasets or EHR datasets may lack.

When a registry does not capture all the necessary information to answer the question of interest in an interventional or non-interventional study, sponsors may consider obtaining supplemental information from another source. If a registry is to be populated with data from another data system, sponsors should consider the potential impact of the additional data on overall integrity of the registry data, in addition to the potential change in overall classification of evidence.

Sponsors should also consider whether the data sources to be linked are interoperable and support appropriate informatics strategies to ensure data integration. Sponsors should ensure that (1) sufficient testing is conducted to demonstrate interoperability of the linked data systems, (2) the automated electronic transmission of data elements to the registry functions in a consistent and repeatable fashion, and (3) data are accurately, consistently, and completely transmitted. Predefined rules to check for logical consistency and value ranges should be used to confirm that data within a registry were retrieved accurately from a linked data source and that the operational definitions for the linked variables are aligned.

ii. Intensive monitoring schemes

The data collection may be undertaken by personnel who attend routinely ward rounds, where they gather information concerning undesirable or unintended events thought by the attending physician to be (potentially) causally related to the medication.

The major strength of such systems is that the monitors may document important information about the events and exposure to medicinal products and the major limitation is the need to maintain a trained monitoring team over time.

Intensive monitoring may be achieved by reviewing medical records or interviewing patients and/or physicians/pharmacists in a sample of healthcare service sites to ensure complete and accurate data on reported adverse events. Some of the weaknesses of this method are problems with selection bias, small numbers of patients and increased costs. These types of studies are most efficient for those medicinal products used mainly in institutional settings such as hospitals, nursing homes and haemodialysis centres. Institutional settings may have a greater frequency of use for certain products and may provide an infrastructure for dedicated reporting. In addition, automatic detection of abnormal laboratory values from computerized laboratory reports in certain clinical settings may provide an efficient intensive monitoring scheme.

iii. Prescription event monitoring (PEM)

Patients may be identified from electronic prescription data or automated health insurance claims. A follow-up questionnaire is sent to each prescribing physician or patient at pre-specified intervals to obtain outcome information (patient demographics, indication for treatment, duration of therapy (including start date), dosage, clinical events and reasons for discontinuation). PEM tends to be used as a method to study safety directly after product launch.

Limitations of PEM include substantial loss to follow-up, relatively short duration of follow-up, selective sampling, selective reporting and limited scope to study products which are used exclusively in hospitals. On the other hand, there is the opportunity to collect more detailed information on adverse events from a large number of physicians and/or patients.

3.2.3 Administrative/ Claims database studies

Administrative data are used increasingly in health services and clinical research in different countries. They are routinely collected during clinic, hospital, laboratory, or pharmacy visits for administrative purposes. Administrative databases provide easy and cheap access to large numbers of patients over expansive geographic regions. Although these databases were initially designed to reimburse health care services and to track differences in services, they are increasingly being used for epidemiological, effectiveness, and safety outcomes research. However, there are several limitations that must be considered, and critical appraisal of studies that utilize administrative databases is important.

Administrative data sets provide a readily available source of “real-world” health care data on a large population of unselected patients. Administrative databases can serve as useful and inexpensive resources for reliably reported data associated with accepted coding systems, including procedure volumes, length of stay, as well as reliably reported outcomes such as death. Furthermore, administrative data can be used to evaluate health care utilization as well as outcomes that differ by patient demographics or geographical region.

One limitation inherent in administrative data is the reason for their creation. Because they are typically intended for financial and administrative management rather than for research purposes, they may vary in the degree of completeness and accuracy.

3.2.4 Pragmatic clinical trials

Pragmatic clinical trial (PCT) are defined as ‘a study comparing several health interventions among a randomized, diverse population representing clinical practice in a real world setting and measuring a broad range of health outcomes.’

PCTs are focused on evaluating benefits and risks of treatments in patient populations and settings that are more representative of routine clinical practice. To ensure generalizability, PCTs should represent the patients to whom the treatment will be applied, for instance, inclusion criteria may be broader (e.g. allowing co-morbidity, co-medication, wider age range), and the follow-up may be minimized and allow for treatment switching.

i. Large simple trials

Large simple trials are pragmatic clinical trials with minimal data collection narrowly focused on clearly defined outcomes important to patients as well as clinicians. Their large sample size provides adequate statistical power to detect even small differences in effects. Additionally, large simple trials include a follow-up time that mimics routine clinical practice.

Large simple trials are particularly suited when an adverse event is very rare or has a delayed latency (with a large expected attrition rate), when the population exposed to the risk is heterogeneous (e.g. different indications and age groups), when several risks need to be assessed in the same trial or when many confounding factors need to be balanced between treatment groups. In these circumstances, the cost and complexity of a traditional RCT may outweigh its advantages and large simple trials can help keep the volume and complexity of data collection to a minimum.

ii. Randomized database studies

Randomized database studies can be considered as a special form of a large simple trial where patients included in the trial are enrolled in a healthcare system with electronic records.

Eligible patients may be identified and flagged automatically by the software, with the advantage of allowing comparison of included and non-included patients. Database screening or record linkage can be used to detect and measure outcomes of interest otherwise assessed through the normal process of care. Patient recruitment, informed consent and proper documentation of patient information are hurdles that still need to be addressed in accordance with the applicable legislation for RCTs. Randomized database studies attempt to combine the advantages of randomization and observational database studies.

3.3 Ethical and Data Protection Requirements

Applicable ethical approvals and data protection requirements outlined in relevant Abu Dhabi and UAE laws, decrees, policies, standards, or regulatory documents must be followed. These include DOH Standard on Human Subject Research, DOH Standard on Patient Healthcare Data Privacy and other relevant documents.

3.4 Amendments/ Deviations and Communication of study Results

Requirements and administrative steps related to study amendments/ deviations and communication of results outlined in DOH Standard on Human Subject Research and other DOH guidelines on clinical research are applicable.

Additionally, for RWE research findings, results should also be reported to dmp@doh.gov.ae

4.Relevant References Documents

No.	Reference Date	Reference Name	Relation Explanation / Coding / Publication Links
1	January 2020	DOH STANDARD ON HUMAN SUBJECT RESEARCH	https://www.doh.gov.ae/en/resources/standards
2	January 2020	DOH Guidelines for Conducting Clinical Trials with Investigational Products and Medical Devices	https://www.doh.gov.ae/en/resources/guidelines
3	February 2019	ABU DHABI - HEALTHCARE INFORMATION AND CYBER SECURITY STANDARD	https://www.doh.gov.ae/en/resources/standards
4	2019	Federal Law No. (2) of 2019 Concerning the Use of Information and Communication Technology (ICT) in Health Fields	https://www.doh.gov.ae/en/about/law-and-legislations
5	January 2020	DOH POLICY ON DIGITAL HEALTH	https://www.doh.gov.ae/en/resources/policies
6	April 2020	DOH POLICY ON THE ABU DHABI HEATLH INFORMATION EXCHANGE	https://www.doh.gov.ae/en/resources/policies
7	SEP 2020	DOH STANDARD ON PATIENT HEALTHCARE DATA PRIVACY	https://www.doh.gov.ae/en/resources/standards

6 Appendix 1 – Checklist for RWD Study Protocols

(adopted from "ENCePP Checklist for Study Protocols (Revision 4)")[6]

Study title:

Study reference number (if applicable):

Section 1: Milestones	**Yes**	**No**	**N/A**	**Section Number**
1.1 Does the protocol specify timelines for				
1.1.1 Start of data collection	☐	☐	☐	
1.1.2 End of data collection	☐	☐	☐	
1.1.3 Progress report(s)	☐	☐	☐	
1.1.4 Interim report(s)	☐	☐	☐	
1.1.5 Final report of study results.	☐	☐	☐	

Comments:

Section 2: Research question	**Yes**	**No**	**N/A**	**Section Number**
2.1 Does the formulation of the research question and objectives clearly explain:	☐	☐	☐	
2.1.1 Why the study is conducted? (e.g. to address an important public health concern, a risk identified in the risk management plan, an emerging safety issue)	☐	☐	☐	
2.1.2 The objective(s) of the study?	☐	☐	☐	
2.1.3 The target population? (i.e. population or subgroup to whom the study results are intended to be generalised)	☐	☐	☐	
2.1.4 Which hypothesis(-es) is (are) to be tested?	☐	☐	☐	
2.1.5 If applicable, that there is no *a priori* hypothesis?	☐	☐	☐	

Comments:

Section 3: Study design	**Yes**	**No**	**N/A**	**Section Number**
3.1 Is the study design described? (e.g. cohort, case-control, cross-sectional, other design)	☐	☐	☐	
3.2 Does the protocol specify whether the study is based on primary, secondary or combined data collection?	☐	☐	☐	

3.3 Does the protocol specify measures of occurrence? (e.g., rate, risk, prevalence)	☐	☐	☐	
3.4 Does the protocol specify measure(s) of association? (e.g. risk, odds ratio, excess risk, rate ratio, hazard ratio, risk/rate difference, number needed to harm (NNH))	☐	☐	☐	
3.5 Does the protocol describe the approach for the collection and reporting of adverse events/adverse reactions? (e.g. adverse events that will not be collected in case of primary data collection)	☐	☐	☐	

Comments:

<u>Section 4: Source and study populations</u>	**Yes**	**No**	**N/A**	**Section Number**
4.1 Is the source population described?	☐	☐	☐	
4.2 Is the planned study population defined in terms of:				
4.2.1 Study time period	☐	☐	☐	
4.2.2 Age and sex	☐	☐	☐	
4.2.3 Country of origin	☐	☐	☐	
4.2.4 Disease/indication	☐	☐	☐	
4.2.5 Duration of follow-up	☐	☐	☐	
4.3 Does the protocol define how the study population will be sampled from the source population? (e.g. event or inclusion/exclusion criteria)	☐	☐	☐	

Comments:

<u>Section 5: Exposure definition and measurement</u>	**Yes**	**No**	**N/A**	**Section Number**
5.1 Does the protocol describe how the study exposure is defined and measured? (e.g. operational details for defining and categorizing exposure, measurement of dose and duration of drug exposure)	☐	☐	☐	
5.2 Does the protocol address the validity of the exposure measurement? (e.g. precision, accuracy, use of validation sub-study)	☐	☐	☐	
5.3 Is exposure categorized according to time windows?	☐	☐	☐	
5.4 Is intensity of exposure addressed? (e.g. dose, duration)	☐	☐	☐	
5.5 Is exposure categorized based on biological mechanism of action and taking into account the pharmacokinetics and pharmacodynamics of the drug?	☐	☐	☐	

5.6 Is (are) (an) appropriate comparator(s) identified?	☐	☐	☐	

Comments:

Section 6: Outcome definition and measurement	**Yes**	**No**	**N/A**	**Section Number**
6.1 Does the protocol specify the primary and secondary (if applicable) outcome(s) to be investigated?	☐	☐	☐	
6.2 Does the protocol describe how the outcomes are defined and measured?	☐	☐	☐	
6.3 Does the protocol address the validity of outcome measurement? (e.g. precision, accuracy, sensitivity, specificity, positive predictive value, use of validation sub-study)	☐	☐	☐	
6.4 Does the protocol describe specific outcomes relevant for Health Technology Assessment? (e.g. HRQoL, QALYs, DALYS, health care services utilization, burden of disease or treatment, compliance, disease management)	☐	☐	☐	

Comments:

Section 7: Bias	**Yes**	**No**	**N/A**	**Section Number**
7.1 Does the protocol address ways to measure confounding? (e.g. confounding by indication)	☐	☐	☐	
7.2 Does the protocol address selection bias? (e.g. healthy user/adherer bias)	☐	☐	☐	
7.3 Does the protocol address information bias? (e.g. misclassification of exposure and outcomes, time-related bias)	☐	☐	☐	

Comments:

Section 8: Effect measure modification	**Yes**	**No**	**N/A**	**Section Number**
8.1 Does the protocol address effect modifiers? (e.g. collection of data on known effect modifiers, sub-group analyses, anticipated direction of effect)	☐	☐	☐	

Comments:

Section 9: Data sources	**Yes**	**No**	**N/A**	**Section Number**
9.1 Does the protocol describe the data source(s) used in the study for the ascertainment of:				

9.1.1 Exposure? (e.g. pharmacy dispensing, general practice prescribing, claims data, self-report, face-to-face interview)	☐	☐	☐	
9.1.2 Outcomes? (e.g. clinical records, laboratory markers or values, claims data, self-report, patient interview including scales and questionnaires, vital statistics)	☐	☐	☐	
9.1.3 Covariates and other characteristics?	☐	☐	☐	
9.2 Does the protocol describe the information available from the data source(s) on:				
9.2.1 Exposure? (e.g. date of dispensing, drug quantity, dose, number of days of supply prescription, daily dosage, prescriber)	☐	☐	☐	
9.2.2 Outcomes? (e.g. date of occurrence, multiple event, severity measures related to event)	☐	☐	☐	
9.2.3 Covariates and other characteristics? (e.g. age, sex, clinical and drug use history, co-morbidity, co-medications, lifestyle)	☐	☐	☐	
9.3 Is a coding system described for:				
9.3.1 Exposure? (e.g. WHO Drug Dictionary, Anatomical Therapeutic Chemical (ATC) Classification System)	☐	☐	☐	
9.3.2 Outcomes? (e.g. International Classification of Diseases (ICD), Medical Dictionary for Regulatory Activities (MedDRA))	☐	☐	☐	
9.3.3 Covariates and other characteristics?	☐	☐	☐	
9.4 Is a linkage method between data sources described? (e.g. based on a unique identifier or other)	☐	☐	☐	

Comments:

Section 10: Analysis plan	**Yes**	**No**	**N/A**	**Section Number**
10.1 Are the statistical methods and the reason for their choice described?	☐	☐	☐	
10.2 Is study size and/or statistical precision estimated?	☐	☐	☐	
10.3 Are descriptive analyses included?	☐	☐	☐	
10.4 Are stratified analyses included?	☐	☐	☐	
10.5 Does the plan describe methods for analytic control of confounding?	☐	☐	☐	
10.6 Does the plan describe methods for analytic control of outcome misclassification?	☐	☐	☐	
10.7 Does the plan describe methods for handling missing data?	☐	☐	☐	
10.8 Are relevant sensitivity analyses described?	☐	☐	☐	

Comments:

Section 11: Data management and quality control	**Yes**	**No**	**N/A**	**Section Number**
11.1 Does the protocol provide information on data storage? (e.g. software and IT environment, database maintenance and anti-fraud protection, archiving)	☐	☐	☐	
11.2 Are methods of quality assurance described?	☐	☐	☐	
11.3 Is there a system in place for independent review of study results?	☐	☐	☐	

Comments:

Section 12: Limitations	**Yes**	**No**	**N/A**	**Section Number**
12.1 Does the protocol discuss the impact on the study results of:				
12.1.1 Selection bias?	☐	☐	☐	
12.1.2 Information bias?	☐	☐	☐	
12.1.3 Residual/unmeasured confounding? (e.g. anticipated direction and magnitude of such biases, validation sub-study, use of validation and external data, analytical methods).	☐	☐	☐	
12.2 Does the protocol discuss study feasibility? (e.g. study size, anticipated exposure uptake, duration of follow-up in a cohort study, patient recruitment, precision of the estimates)	☐	☐	☐	

Comments:

Section 13: Ethical/data protection issues	**Yes**	**No**	**N/A**	**Section Number**
13.1 Have requirements of Ethics Committee/ Institutional Review Board been described?	☐	☐	☐	
13.2 Has any outcome of an ethical review procedure been addressed?	☐	☐	☐	
13.3 Have data protection requirements been described?	☐	☐	☐	

Comments:

<u>Section 14: Amendments and deviations</u>	**Yes**	**No**	**N/A**	**Section Number**
14.1 Does the protocol include a section to document amendments and deviations?	☐	☐	☐	

Comments:

<u>Section 15: Plans for communication of study results</u>	**Yes**	**No**	**N/A**	**Section Number**
15.1 Are plans described for communicating study results (e.g. to regulatory authorities)?	☐	☐	☐	
15.2 Are plans described for disseminating study results externally, including publication?	☐	☐	☐	

Comments:

Name of the main author of the protocol: ______________________________

Date: dd/Month/year

Signature: ________________________

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